

IMPACT OF 6G NETWORK PERFORMANCE ON MANUFACTURING EFFICIENCY IN SMART FACTORIES

A Comprehensive Data Science Research Report

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I. INTRODUCTION & INDUSTRY CONTEXT

The global smart manufacturing and Industry 4.0 sectors are experiencing an unprecedented digital transformation. Smart factories increasingly combine connected machines, robotics, industrial automation, sensors, and real-time analytics to improve production efficiency.

Within these environments, communication quality is closely connected to the ability of machines and control systems to exchange information in a timely and reliable manner. Ultra-reliable low-latency communication is therefore an important consideration for advanced industrial networking.

This project investigates the impact of 6G network performance on manufacturing efficiency by analyzing network latency and packet loss alongside production speed, quality-control defect rate, and overall efficiency status. The study reframes factory efficiency as a combined manufacturing-and-network performance problem rather than a purely machine-output problem.

The analytical objective is to develop measurable indicators and an interactive monitoring framework that can help engineering stakeholders identify degraded communication conditions, understand their relationship with production outcomes, and support proactive operational action.

II. DATASET DESCRIPTION & PRE-PROCESSING METHODOLOGY

The analytical foundation of this project is the **Thales_Group_Manufacturing.csv** dataset, described in the supplied implementation as containing 100,000 distinct operational telemetry records from advanced smart-factory floors.

A. Data Ingestion and Inspection

The raw dataset is ingested using the Pandas library in Python. The implementation first displays the first five records, inspects dataset information and columns, and checks the dataset for missing values.

B. Validation of Key Variables

The project validates the principal network and manufacturing fields used throughout the analysis:

Network_Latency_ms, Packet_Loss_%, Production_Speed_units_per_hr,
Quality_Control_Defect_Rate_%, and Efficiency_Status.

C. KPI Engineering

Two additional variables are generated. Network_Stability_Index applies normalized penalties to latency and packet loss. Latency_Sensitivity_Score divides production speed by latency plus one, creating a compact indicator of production speed relative to network delay.

D. Processed Dataset

After KPI calculation, the enhanced dataset is exported as

Thales_Group_Manufacturing_Processed.csv, which is subsequently used by the Streamlit dashboard.

III. THE 6G MANUFACTURING PERFORMANCE FRAMEWORK

Moving beyond one-dimensional analysis, this project organizes factory operating conditions into four actionable performance profiles. The framework links communication conditions with production behavior and provides a basis for targeted engineering response.

Profile Name	Criteria Definition	Operational Characteristics & Strategic Implication
Optimal High-Speed	Latency < 15 ms with zero packet loss.	Maximum production speed and strong machine synchronization. Strategy: maintenance and continuous stability monitoring.
At-Risk Latency	Moderate latency from 15-45 ms.	Potential throughput throttling and synchronization delays. Strategy: preemptive edge-routing and connectivity optimization.
Critical Degradation	High packet loss with latency > 45 ms.	Severe communication degradation with potential production bottlenecks and quality-control deterioration. Strategy: immediate automated network failover.
Low-Impact Idle	Low operational speed during maintenance.	Natural downtime with comparatively limited impact on active production output. Strategy: standard offline/maintenance protocol.

IV. IN-DEPTH KEY PERFORMANCE INDICATOR (KPI) ANALYSIS

The project formulates multiple KPIs to quantify network health and its relationship with manufacturing efficiency. Each KPI provides a distinct analytical lens.

A. Network Stability Index

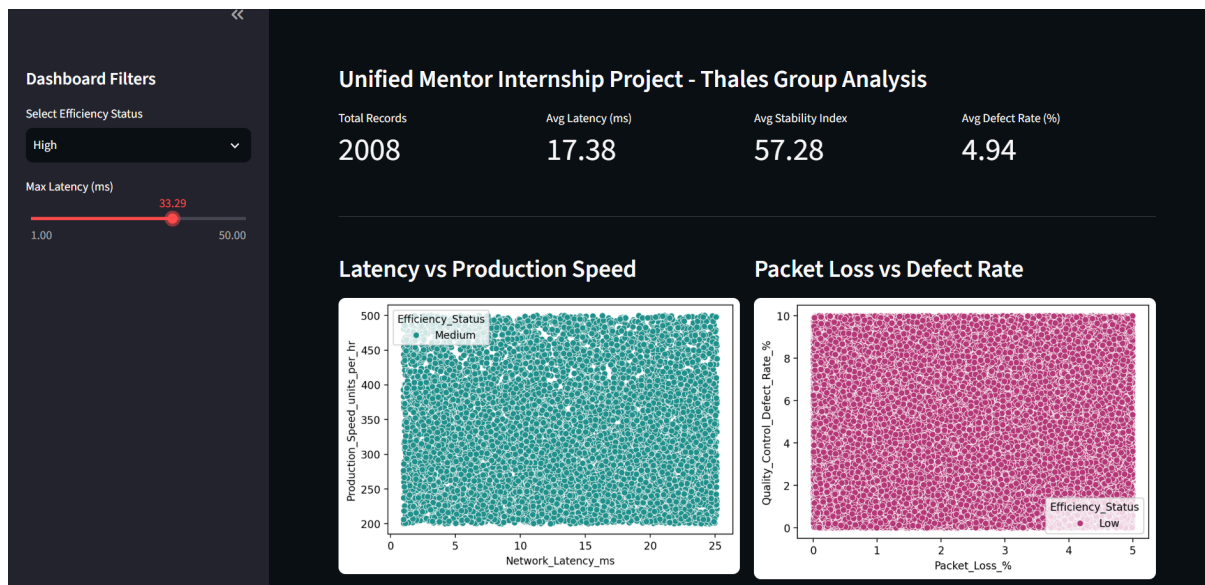
The Network Stability Index combines normalized network latency and packet loss on an inverse scale. The implementation assigns equal 50-point contributions to normalized latency and packet loss penalties, producing a 0–100-style stability measure.

B. Latency Sensitivity Score

This metric is calculated as production speed divided by network latency plus one. It provides an operational measure of production speed relative to communication delay and helps identify conditions where production is more sensitive to latency.

C. Packet Loss & Quality-Control Defect Relationship

The project visualizes Packet_Loss_% against Quality_Control_Defect_Rate_% to inspect whether increasing communication loss is associated with deterioration in quality-control outcomes.



IV. IN-DEPTH KEY PERFORMANCE INDICATOR (KPI) ANALYSIS — CONTINUED

D. Network Latency & Production Speed Relationship

The latency-versus-production-speed scatter analysis uses Efficiency_Status as the grouping variable. This allows observations to be examined according to their operational efficiency category and provides a direct visual connection between network delay and production throughput.

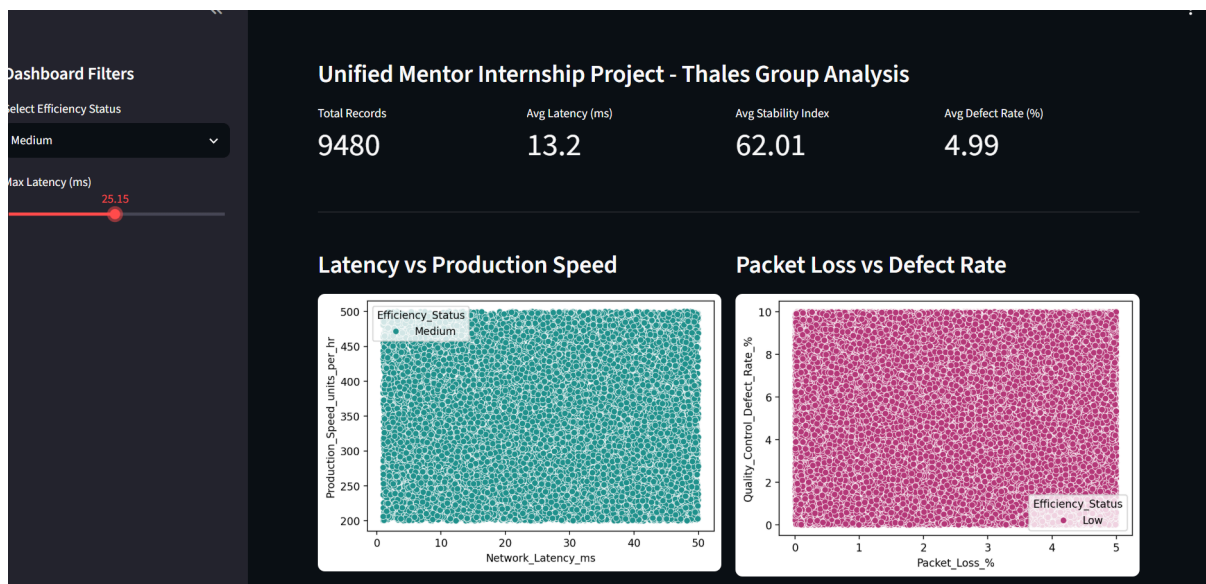
E. Efficiency Status as an Operational Ground Truth

Efficiency_Status is retained as the principal categorical reference for interpreting the network-performance relationships. It enables the dashboard to compare records under different operating conditions and supports filtering for engineering review.

F. Analytical Threshold Framework

The implementation uses three principal network-performance bands: below 15 ms for optimal high-speed conditions; 15–45 ms for at-risk latency; and above 45 ms, when accompanied by high packet loss, for critical degradation. These thresholds are used as project classification criteria rather than universal industrial standards.

Taken together, these KPIs convert raw network telemetry into interpretable indicators that can be connected to production and quality outcomes. This creates a bridge between network engineering and manufacturing-performance management.



V. TECHNICAL IMPLEMENTATION: STREAMLIT WEB APPLICATION ARCHITECTURE

To operationalize the analytical findings and transition the project from a static analysis into a dynamic engineering tool, a Streamlit web application was designed. The application uses the processed dataset containing the engineered KPIs.

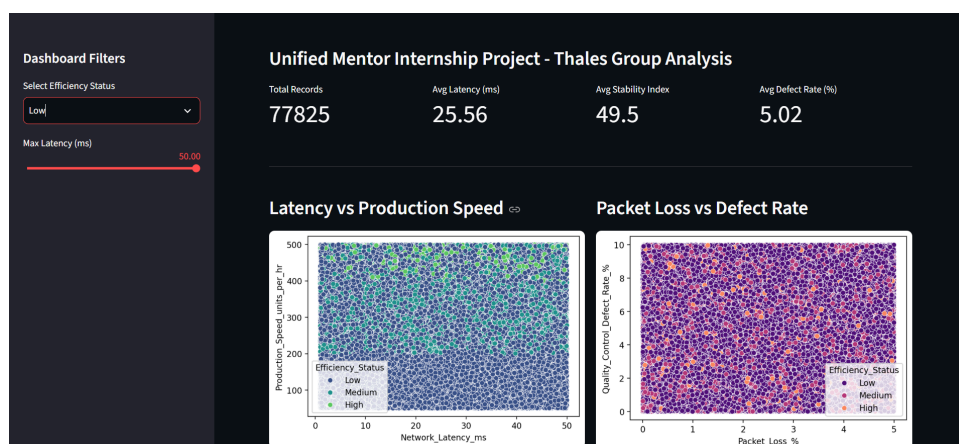
Technology Stack: Python, Pandas, NumPy, Streamlit, Matplotlib, and Seaborn are used across the data-processing, KPI, visualization, and dashboard layers.

Interactive Core Modules: The dashboard includes KPI cards for total records, average latency, average Network Stability Index, and average defect rate. It also presents the Latency vs Production Speed and Packet Loss vs Defect Rate visualizations.

User Capabilities & Filtering: Stakeholders can filter the dataset by Efficiency Status and adjust a maximum network-latency threshold through the dashboard sidebar.

Operational Monitoring: The dashboard is intended to function as a continuous analytical interface for engineering teams, allowing network conditions and manufacturing outcomes to be reviewed together rather than independently.

Dashboard KPI	Purpose
Total Records	Shows the number of records remaining after the selected filters.
Average Latency (ms)	Summarizes communication delay for the filtered operating conditions.
Average Stability Index	Summarizes the combined latency and packet-loss stability measure.
Average Defect Rate (%)	Summarizes the quality-control defect rate for the filtered records.



VI. STRATEGIC RECOMMENDATIONS

Phase 1: Immediate Interventions (0–3 Months)

Deploy automated alerts for network-latency and packet-loss threshold breaches. When a factory segment enters an at-risk or critical condition, engineering teams should receive an actionable notification so that connectivity issues can be reviewed before they propagate into production disruptions.

Phase 2: Engineering Optimization

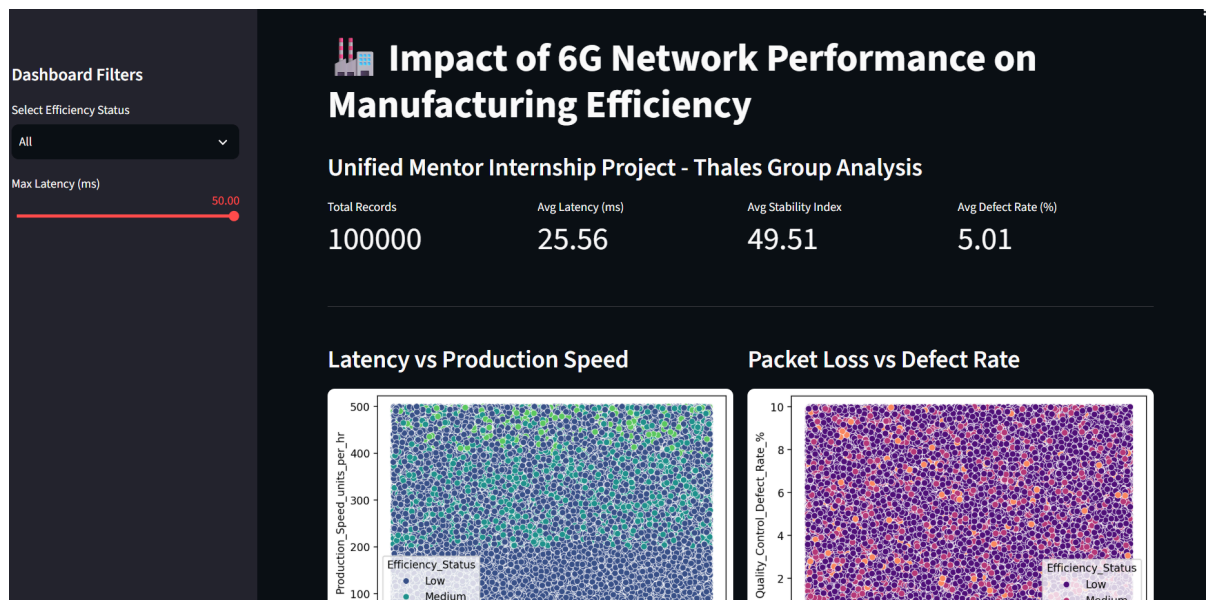
Use the latency and packet-loss analysis to review edge-routing, local wireless infrastructure, and communication paths. The objective is to reduce avoidable delay and improve reliability for time-sensitive manufacturing processes.

Phase 3: Long-Term Network Modernization (6+ Months)

Evaluate upgrades to industrial wireless and edge-network infrastructure capable of supporting ultra-reliable, low-latency communication requirements. Network modernization should be evaluated alongside measured production and quality outcomes.

Continuous Monitoring via Dashboard

Embed the Streamlit analytics dashboard into the regular workflow of engineering and production teams. Repeated monitoring of latency, stability, production speed, and defect rate can help identify changes in operational conditions over time.



VII. CONCLUSION

The project demonstrates a structured approach for evaluating the relationship between 6G-oriented network performance and manufacturing efficiency in smart-factory environments. Rather than treating network telemetry as an isolated technical concern, the analysis places latency and packet loss alongside production speed and quality-control outcomes.

The formulation of the Network Stability Index and Latency Sensitivity Score provides a compact way to interpret network conditions in relation to factory performance. The latency-versus-production-speed and packet-loss-versus-defect-rate analyses further provide direct visual mechanisms for examining operational relationships.

The four-part performance framework—Optimal High-Speed, At-Risk Latency, Critical Degradation, and Low-Impact Idle—translates these measurements into practical operating categories. The framework is intended to support differentiated engineering responses rather than a one-size-fits-all approach to network management.

Finally, the Streamlit implementation bridges the gap between data analysis and practical monitoring. By bringing raw telemetry, engineered KPIs, filters, and visualizations into a single interactive interface, the project establishes a reusable foundation for data-driven smart-factory network and production oversight.

Overall, the supplied project implementation positions network-aware analytics as an important component of smart-manufacturing efficiency management, while recognizing that the measured relationships and thresholds are specific to the supplied dataset and project methodology.